

UNDERSTANDING THE FINE-GRAINED KNOWLEDGE CAPABILITIES OF VISION-LANGUAGE MODELS

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ABSTRACT

Vision-language models (VLMs) have advanced rapidly across the full spectrum of visual question answering, encompassing visual reasoning, document understanding, and multimodal dialogue, and these gains hold consistently across VLMs built on diverse base language models, alignment architectures, and training corpora. This progress, however, conceals a systematic blind spot: recent evidence establishes that such models lag markedly on traditional image classification benchmarks, precisely the setting that probes the fine-grained visual knowledge underpinning vision-centric competence. In this work, we conduct a comprehensive evaluation of a large and representative set of recent VLMs on fine-grained classification benchmarks and isolate the factors responsible for the disconnect between fine-grained knowledge and performance on other vision benchmarks. Through a systematic series of controlled ablation experiments, we establish that a stronger base LLM raises all benchmark scores in equal measure, whereas a stronger vision encoder disproportionately improves fine-grained classification. We further demonstrate that the pretraining stage is decisive for fine-grained performance, and especially so when the language model weights are unfrozen during pretraining. Collectively, these findings chart a concrete path toward VLMs with substantially stronger fine-grained visual understanding and vision-centric capabilities.

1 INTRODUCTION

Vision-language models (VLMs), which couple a vision encoder with a large language model (LLM), have rapidly acquired increasingly sophisticated capabilities for interpreting and reasoning about visual content (Alayrac et al., 2022; Team et al., 2023; OpenAI, 2023; Liu et al., 2023; Wang et al., 2024b), and now handle demanding tasks that include open-ended visual question answering, document understanding, and multimodal dialogue (Goyal et al., 2017; Yue et al., 2024; Mathew et al., 2021; OpenAI, 2023). This trajectory of progress, however, leaves a foundational question unresolved and largely unexamined: how well do these models perform on fine-grained visual perception, and what concretely can be done to improve it? The present paper takes up this question directly, treating fine-grained recognition not as a peripheral curiosity but as a core axis of visual intelligence.

Fine-grained visual recognition—the ability to discriminate among visually similar categories—demands that a model resolve subtle discriminative features while remaining invariant to intra-class variation in pose, illumination, and background. Traditional vision encoders such as CLIP (Radford et al., 2021) and DINO (Caron et al., 2021) attain strong performance on fine-grained classification benchmarks; the fine-grained behavior of the newer generation of VLMs built atop these very encoders, by contrast, has gone comparatively unexamined.

Characterizing the fine-grained knowledge capabilities of VLMs is essential precisely because perception constitutes the foundation on which higher-order understanding and reasoning are built: a model cannot reason reliably about what it cannot first recognize. Real-world deployments of VLMs routinely confront fine-grained, long-tailed distributions in which discriminating among closely related categories is decisive rather than incidental. Should a model misidentify a mushroom species, for example, no amount of downstream textual knowledge can recover whether the specimen is poisonous or safe to consume. Beyond such immediate stakes, a principled account of the factors that

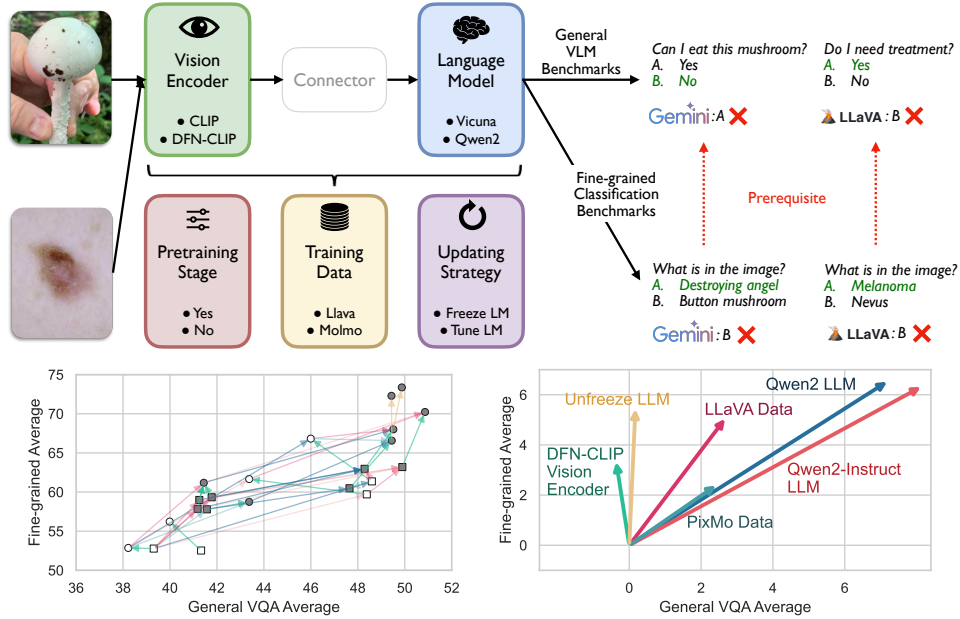


Figure 1: **Overview.** (Top) We investigate the fine-grained classification capabilities of vision-language models (VLMs)—a foundational yet routinely overlooked dimension of visual competence that underpins higher-level understanding and reasoning. (Bottom) Across 22 systematic ablation experiments spanning the principal model components and training strategies (left), we isolate the factors that govern fine-grained classification performance in VLMs (right).

govern fine-grained knowledge acquisition directly informs the design of more effective architectures and training strategies for the next generation of VLMs.

We first present a comprehensive evaluation of fine-grained visual knowledge in state-of-the-art VLMs, directly juxtaposing their performance on fine-grained classification benchmarks against their capabilities on general VLM tasks. Concretely, we evaluate 15 representative VLMs—spanning LLaVA (Liu et al., 2023; 2024a), Phi (Abdin et al., 2024), Qwen2-VL (Wang et al., 2024b), and Molmo (Deitke et al., 2024)—on four fine-grained classification benchmarks that together cover object, flower, pet, and food domains: ImageNet (Deng et al., 2009), Flowers (Nilsback & Zisserman, 2008), Pets (Parkhi et al., 2012), and Food (Bossard et al., 2014), alongside aggregated results from eight general VLM benchmarks including MMMU (Yue et al., 2024), MathVista (Lu et al., 2024), and MMVet (Yu et al., 2024). This analysis establishes that models with comparable general ability nonetheless diverge substantially in fine-grained classification accuracy, demonstrating that fine-grained recognition constitutes a distinct axis of visual intelligence that current VLM benchmarks fail to capture. It further reveals a persistent and consistent gap between VLMs and the CLIP-based vision encoders they are built upon, corroborating and extending prior observations (Zhang et al., 2024; Geigle et al., 2024).

To pinpoint what drives fine-grained classification performance in VLMs, we then conduct a systematic study that disentangles the contributions of the principal model components and training paradigms. Specifically, we quantify how the two core architectural components—the language model and the vision encoder—and the training strategy—pretraining, the choice of which weights are updated, and data quality—each shape fine-grained classification. Through 22 carefully controlled ablation experiments, we identify the factors that decisively govern fine-grained knowledge acquisition in VLMs:

- **Language model (§4.2):** Upgrading the base language model raises performance uniformly, yielding comparable gains on fine-grained classification and general VLM benchmarks alike.
- **Vision encoder (§4.2):** A stronger vision encoder selectively improves fine-grained classification while exerting only limited influence on general VLM benchmarks, an asymmetry that is most pronounced in models trained in the standard two-stage regime.

- **Pretraining stage (§4.3):** Large-scale pretraining on image-captioning data delivers substantial gains in fine-grained classification, with a comparatively modest effect on general VLM benchmarks.
- **Weight updating method (§4.3):** Jointly updating the vision-encoder-to-LLM connector and the LLM during pretraining improves fine-grained classification well beyond what connector-only pretraining achieves.
- **Data quality (§4.3):** The quality of the pretraining and instruction-tuning data exerts only a limited effect on fine-grained classification, an effect that all but vanishes when the language model weights are held frozen during pretraining.

In sum, this study places fine-grained visual perception in VLMs—an essential capability that current benchmark evaluations systematically fail to surface—at the center of analysis, and through controlled ablations identifies the factors that decisively govern fine-grained knowledge acquisition. Taken together, these findings supply actionable guidance for building vision-centric VLMs with markedly stronger visual understanding, and thereby for improving their reliability on the real-world long-tail tasks that hinge on fine-grained recognition.

2 RELATED WORK

Vision-language models and their evaluation. Vision-language models (VLMs) integrate visual inputs with a language model and predominantly adopt the LLaVA-style architecture (Liu et al., 2023), which comprises three core components: a vision encoder, a language model, and an MLP connector that bridges the two. Such models attain strong performance across a broad range of multimodal tasks, including visual question answering (VQA), document understanding, and general reasoning (Liu et al., 2023; Dai et al., 2023; Awadalla et al., 2023; Wang et al., 2024b). Their evaluation, correspondingly, rests on a large family of benchmarks such as MMMU (Yue et al., 2024), MathVista (Wang et al., 2024a), and DocVQA (Mathew et al., 2021). These benchmarks, however, predominantly probe reasoning and language understanding conditioned on visual input, while leaving core vision-centric competencies—object recognition and fine-grained classification foremost among them—essentially unmeasured. We close this gap by systematically evaluating and analyzing VLMs on fine-grained classification, an essential yet underexplored dimension of visual intelligence.

Fine-grained visual classification. Fine-grained classification is a well-established computer vision task that seeks to discriminate among visually similar subcategories within a broader object category, such as bird species (Wah et al., 2011), flowers (Nilsback & Zisserman, 2008), and pet breeds (Parkhi et al., 2012). The corresponding benchmarks have long served as central testbeds for training and evaluating vision encoders including ResNet (He et al., 2016), CLIP (Radford et al., 2021), and DINO (Caron et al., 2021). Despite this foundational role in vision research, recent VLMs have largely omitted these datasets from their evaluation protocols in favor of visual question answering. Because fine-grained classification is indispensable to real-world applications that involve long-tail distributions and demand precise recognition—medical diagnosis, food safety, and species identification among them—evaluating VLMs on such benchmarks is essential for assessing their practical usability from a vision-centric perspective, and not merely on language-centric reasoning tasks.

Evaluation and ablation of VLMs for fine-grained classification. Prior studies of VLM evaluation on fine-grained classification (Zhang et al., 2024; Geigle et al., 2024) remain limited in scope, typically confined to a handful of models or datasets. We substantially broaden this line of work, evaluating a considerably wider range of VLMs across multiple fine-grained classification benchmarks within a single unified multiple-choice protocol. Beyond evaluation, we contribute a comprehensive ablation analysis that isolates the key factors driving fine-grained classification performance—an analysis without precedent in this setting.

A parallel body of work examines the design space of VLMs more generally, dissecting model architectures (McKinzie et al., 2024), training strategies (Karamcheti et al., 2024), and the role of multimodal datasets (Gadre et al., 2023; Udandarao et al., 2024; Deitke et al., 2024). These studies, however, are oriented almost exclusively toward general benchmarks that emphasize language and reasoning. We instead trace how the central design choices—base model selection, pre-training methodology, and the composition of the pretraining and instruction-tuning data—shape

fine-grained classification specifically, yielding novel and directly actionable insights for optimizing VLMs on vision-centric tasks.

3 EVALUATION: FINE-GRAINED CLASSIFICATION

This section motivates the evaluation of fine-grained classification, specifies our benchmarks and evaluation protocol, and reports the central observations that emerge from testing 15 existing VLMs.

3.1 MOTIVATION

Fine-grained classification is a foundational computer vision task that targets the discrimination of visually similar subcategories within a broader category—for instance, distinct bird species or car models (Wei et al., 2021). A substantial suite of evaluation benchmarks, ImageNet (Deng et al., 2009) and Flowers (Nilsback & Zisserman, 2008) chief among them, has become the standard testbed for training and evaluating vision encoders. With the advent of large-scale self-supervised and supervised learning, vision encoders such as CLIP (Radford et al., 2021) and DINO (Caron et al., 2021) attain near- or super-human accuracy even on the most challenging of these fine-grained tasks.

These benchmarks, however, have been conspicuously absent from the evaluation of vision-language models (VLMs). Although recent VLMs such as LLaVA (Liu et al., 2023) demonstrate strong capabilities in interpreting visual concepts, their evaluation remains largely confined to visual question answering (VQA)—academic VQA benchmarks such as MMMU (Yue et al., 2024) and MathVista (Lu et al., 2024), or document understanding tasks such as DocVQA (Mathew et al., 2021) and TextVQA (Singh et al., 2019). Because these benchmarks foreground reasoning and language processing yet contain no fine-grained object categories, they leave the core vision-centric capabilities of VLMs essentially unexamined.

Fine-grained classification is, nonetheless, decisive for VLMs, since real-world applications routinely encounter long-tailed distributions that demand precise visual discrimination among closely related categories. Consider a blind user asking whether a mushroom is safe to eat: the model must first identify the species correctly before it can reason about toxicity at all, and misclassifying a deadly destroying angel as an edible white button mushroom could prove fatal. The same logic holds in medical diagnosis, where a failure to distinguish visually similar conditions can precipitate incorrect treatment and serious harm, and in autonomous driving, where confusing a “stop” sign with a “do not enter” sign—both red and similarly shaped—can produce dangerous navigation errors. Excelling at fine-grained classification is therefore not a peripheral desideratum but a prerequisite for the safety and reliability of VLMs in real-world deployment.

3.2 BENCHMARKS

We probe the fine-grained visual knowledge of VLMs with four well-established object recognition benchmarks, each targeting a distinct domain of fine-grained classification and together spanning generic objects, flowers, pets, and food: **ImageNet-1K** (Deng et al., 2009), **Oxford Flowers-102** (Nilsback & Zisserman, 2008), **Oxford-IIIT Pet-37** (Parkhi et al., 2012), and **Food-101** (Bossard et al., 2014).

Because these benchmarks span 37 to 1,000 distinct classes, evaluating VLMs directly in the multiple-choice format that is standard for VLM training and evaluation is not straightforward. We resolve this by adopting the methodology of Geigle et al. (2024) to recast each dataset as a set of 5-way multiple-choice questions. Concretely, for every image we use an OpenCLIP ViT-L/14 model (Schuhmann et al., 2022) to retrieve the four hardest negative choices by image-label cosine similarity, and combine them with the ground-truth label to form the answer set. We deliberately select the OpenCLIP model so as not to bias question difficulty against VLMs that rely on OpenAI-CLIP (Radford et al., 2021), SigLIP (Zhai et al., 2023), or DFN-CLIP (Fang et al., 2023), each of which is trained on a corpus distinct from OpenCLIP’s. This construction lets us evaluate VLMs in their native format while preserving the bulk of the task’s intrinsic difficulty: CLIP ViT-L/14-336px (Radford et al., 2021), for example, attains 75.2% accuracy on the full 102-class Flowers-102 and rises to only 78.3% on our 5-way multiple-choice version, confirming that the reformulation retains most of the discriminative challenge.

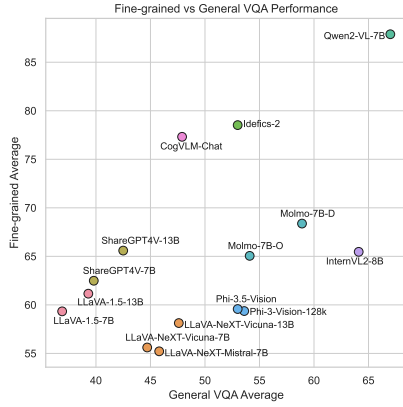


Figure 2: **Fine-grained classification versus general VQA performance across VLMs.** Across recent VLMs, fine-grained classification accuracy varies widely at fixed general VQA performance, establishing fine-grained classification as a distinct axis of visual capability that standard VQA benchmarks do not measure.

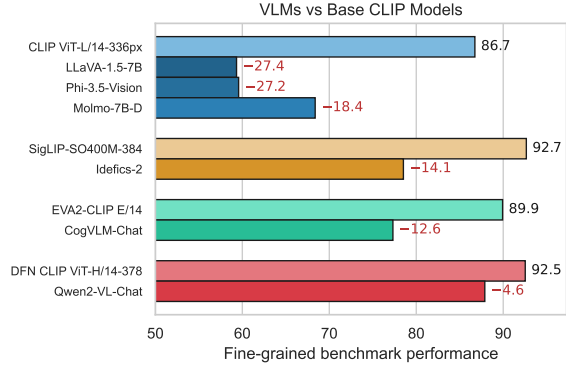


Figure 3: **VLMs versus their own CLIP vision encoders on fine-grained classification.** Although Qwen2-VL-Chat nearly recovers the fine-grained accuracy of its DFN-CLIP vision encoder, every other VLM trails its own encoder by a wide margin, quantifying the substantial fine-grained capability that current VLMs leave unrealized.

3.3 TESTING SETUP

We evaluate 15 recent VLMs spanning the 7B–13B parameter range on the converted 5-choice fine-grained classification benchmarks (Liu et al., 2023; 2024a; Chen et al., 2024a; Abidin et al., 2024; Deitke et al., 2024; Laurençon et al., 2024; Hong et al., 2024; Chen et al., 2024c; Wang et al., 2024b). For each VLM, we adopt its respective default prompt format from VLMEvalKit (Duan et al., 2024) and score accuracy by exact match against the multiple-choice options. As a baseline, we evaluate a range of CLIP models (Radford et al., 2021; Schuhmann et al., 2022; Fang et al., 2023) under the standard zero-shot protocol, in which both the image and each candidate class label (“A photo of a [class]”) are mapped through CLIP’s image and text encoders and the top-1 prediction is taken as the label of maximal image–text cosine similarity.

To determine whether existing VLM benchmarks already capture fine-grained classification ability, we compare each VLM’s fine-grained accuracy against its average score on eight general VQA benchmarks from VLMEvalKit, including MMMU (Yue et al., 2024) and MathVista (Lu et al., 2024).

3.4 RESULTS

Figure 2 plots fine-grained classification performance (averaged across our four benchmarks) against general VQA performance (averaged across eight existing VQA benchmarks) for every tested VLM. The two quantities decouple sharply: models with essentially identical general VQA performance span a wide range of fine-grained classification accuracy. CogVLM-Chat (Hong et al., 2024) and LLaVA-NeXT-Vicuna-13B (Liu et al., 2024a), for instance, both reach nearly 48% average accuracy on general VQA, yet their fine-grained classification accuracy differs by a full 19 percentage points (pp)—77.3% for CogVLM-Chat against 58.1% for LLaVA-NeXT-Vicuna-13B. A gap of this magnitude between models matched on general VQA demonstrates that fine-grained classification ability is not captured by existing VLM benchmarks, and establishes it as a distinct dimension of visual intelligence.

Finding 1: Fine-grained classification represents a distinct aspect of visual capability that current VLM benchmarks fail to adequately measure.

Every current VLM is constructed on top of a pretrained vision encoder, and our multiple-choice reformulation exploits this fact to enable a direct, like-for-like comparison between each VLM and the CLIP model it inherits on traditional classification benchmarks. Figure 3 contrasts the fine-grained classification accuracy of selected VLMs with that of their corresponding CLIP baselines, exposing a substantial and consistent shortfall. Molmo (Deitke et al., 2024), a very recent model that excels across VQA benchmarks, attains only 68.4% average accuracy in our multiple-choice setting, whereas its own CLIP ViT-L/14 (Radford et al., 2021) vision encoder reaches 86.7%—a gap of 18.3pp. Even the strongest model we test, Qwen2-VL-Chat (Wang et al., 2024b), trails its own state-of-the-art DFN-CLIP (Fang et al., 2023) vision encoder by 4.6 points. That fine-grained knowledge already present in the vision encoder is not fully preserved through multimodal integration motivates a systematic analysis of VLM design choices to identify the factors that recover and enhance fine-grained classification. We undertake precisely this ablation study in the next section.

Finding 2: A significant performance gap exists between VLMs and CLIP models in fine-grained classification tasks.

4 ABLATIONS: BASE MODELS AND TRAINING

4.1 EXPERIMENTAL SETUP

Building on the observational results of Section 3, we now ablate the model-level differences that plausibly drive fine-grained classification performance, concentrating on the two dimensions most likely to matter: base model selection and training methodology. All experiments are conducted within the LLaVA-1.5 training framework and alignment architecture (Liu et al., 2023), holding the surrounding pipeline fixed so that each ablation isolates a single factor. Every model undergoes one epoch of pretraining (where applicable) followed by one epoch of finetuning, and we retain LLaVA-1.5’s default hyperparameters—learning rate and batch size included—throughout.

We report fine-grained performance as the average accuracy across the four classification benchmarks introduced in Section 3.2. For general VQA performance, we report the average over three widely used multiple-choice benchmarks—MMM (Yue et al., 2024), MMBench (Liu et al., 2024b), and MMStar (Chen et al., 2024b)—selected for their broad domain coverage and established role in VLM evaluation.

4.2 BASE MODELS

The analysis in Section 3 reveals that recent models have steadily adopted newer base LLMs while, in most cases, retaining the CLIP ViT-L/14 vision encoder of Radford et al. (2021). We therefore isolate how each of these two base-model choices individually shapes fine-grained classification performance.

4.2.1 LLM CHOICE

Setup. Taking LLaVA-1.5-7B (Liu et al., 2023) as our baseline, we vary the LLM component while sweeping across vision encoders and pretraining configurations, so that the LLM effect can be read off independently of the other design axes. We compare the original Vicuna-7B-1.5 (Chiang et al., 2023) used in LLaVA against Qwen2-7B (Yang et al., 2024) (the LLM of Qwen2-VL-7B (Wang et al., 2024b) and Molmo-7B-D (Deitke et al., 2024)), together with Llama-2-7B (Touvron et al., 2023) (Vicuna’s base model) and Qwen2-7B-Instruct (the instruction-tuned variant of Qwen2-7B).

Findings. As Figure 4 shows, replacing Vicuna-7B with Qwen2-7B improves performance substantially and uniformly across every configuration, with average gains of +7.5 percentage points (pp) on fine-grained benchmarks and +7.4pp on general VQA benchmarks. The instruction-tuned Qwen2-7B-Instruct yields closely comparable gains (+7.5pp and +8.1pp, respectively), whereas Llama-2-7B, the non-instruction-tuned base of Vicuna, produces only modest improvements whose downstream implications, absent instruction tuning, remain difficult to interpret. Crucially, all of our trained models—including the weaker Vicuna-based VLMs—emit multiple-choice responses in the correct format essentially without exception. The observed differences in fine-grained accuracy therefore

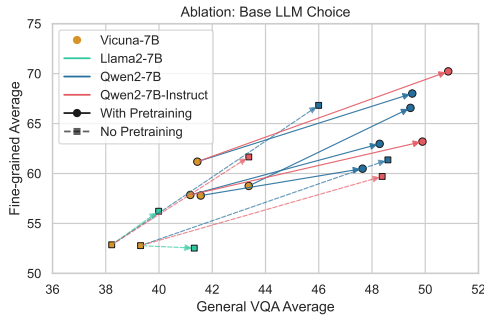


Figure 4: **Base-LLM ablation: replacing LLaVA’s Vicuna-7B with alternative LLMs.** Averaged across vision encoders and pretraining configurations, switching from Vicuna to Qwen2-7B raises fine-grained performance by +7.5pp and general VQA performance by +7.7pp, confirming that a stronger LLM lifts both axes in tandem.

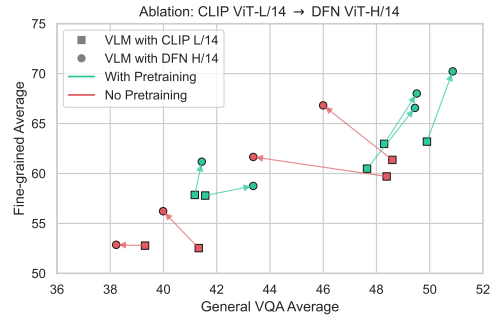


Figure 5: **Base vision-encoder ablation: replacing LLaVA’s CLIP ViT-L/14 with DFN-CLIP ViT-H/14.** Given sufficient pretraining, the stronger DFN-CLIP encoder improves fine-grained performance by +4.5pp while lifting general performance by only +1.2pp, isolating a vision-encoder effect that is specific to fine-grained classification.

reflect genuinely improved knowledge of the presented options rather than mere gains in format adherence, isolating knowledge as the operative mechanism.

Takeaway 1: Stronger language models consistently improve performance across both fine-grained classification and general VQA benchmarks.

4.2.2 VISION ENCODER CHOICE

Setup. Most of the evaluated VLMs, the LLaVA series and Molmo among them, adopt OpenAI-CLIP ViT-L/14-336px (Radford et al., 2021) as their vision encoder, whereas Qwen2-VL (Wang et al., 2024b) employs DFN-CLIP ViT-H/14-378px (Fang et al., 2023), which delivers markedly stronger zero-shot performance on object recognition tasks such as ImageNet. We contrast the two encoders across the full range of LLM choices and pretraining configurations.

Findings. Figure 5 demonstrates that the benefit of substituting CLIP L/14 with DFN-CLIP H/14 is contingent on whether connector pretraining precedes finetuning. Absent pretraining, DFN-CLIP slightly degrades general VQA performance (-2.5pp) while modestly improving fine-grained scores (+2.8pp). Once connector pretraining is in place, however, the picture changes decisively: general VQA performance holds steady or improves marginally (+1.2pp), while fine-grained performance rises substantially (+4.5pp). A stronger vision encoder thus translates into a large, targeted gain in fine-grained classification—provided the encoder is properly integrated into the model through pretraining.

Takeaway 2: Better vision encoders (like DFN-CLIP vs. CLIP) in VLMs improve fine-grained classification, but only when these encoders are properly integrated through pretraining before the finetuning stage.

4.3 TRAINING METHODS

Recent VLMs are trained almost universally in two stages (Liu et al., 2023; Wang et al., 2024b; Chen et al., 2024c). The first stage, pretraining, optimizes a captioning objective on large-scale image-caption data to establish alignment between the vision encoder and the LLM. The second stage, instruction tuning, refines the aligned model on carefully curated multi-round, multimodal instruction-completion data, aligning it with human intent for real-world use. We now quantify how each of these two stages contributes to the fine-grained capabilities of VLMs.

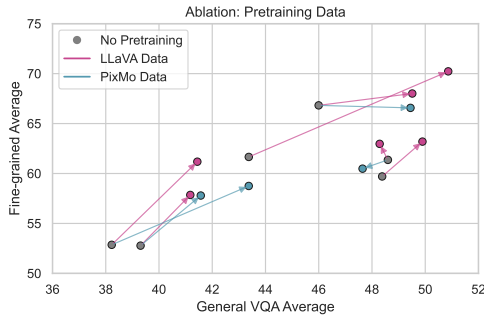


Figure 6: **Pretraining-data ablation.** Relative to no pretraining, adding either LLaVA (CC-3M) or Molmo (PixMo-Cap) connector-pretraining data yields comparable gains: LLaVA data raises fine-grained and general scores by +4.0pp and +2.1pp, and PixMo data by +2.4pp and +2.5pp, respectively—so despite a large disparity in caption quality, the two sources behave almost interchangeably.

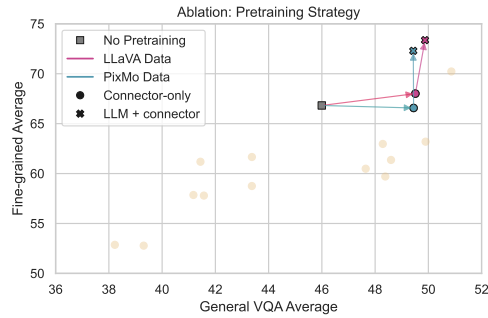


Figure 7: **Training-method ablation: connector-only pretraining versus joint LLM+connector pretraining.** Unfreezing the LLM during pretraining improves fine-grained performance by +5.5pp with no adverse effect on general VQA scores, identifying which weights are updated—not merely whether pre-training occurs—as a key lever for fine-grained classification.

Table 1: **Summary of the principal ablation settings, with existing models bolded for reference.** Upgrading the base LLM and the vision encoder, together with pretraining the connector and LLM on sufficient data, each contributes substantial gains in fine-grained performance, whereas instruction finetuning contributes comparatively little. General VQA benchmarks exhibit the complementary pattern, with the choice of LLM accounting for the largest share of the improvement.

Model/Ablation	Vision Encoder	LLM	Arch.	Pretraining Data	Finetuning Data	Fine-grained Classification	General VQA
LLaVA-1.5-7B	CLIP L/14	Vicuna	LLaVA	LLaVA	LLaVA	59.3	41.8
No pretraining	CLIP L/14	Vicuna	LLaVA	None	LLaVA	52.8	39.3
LLaVA reproduction	CLIP L/14	Vicuna	LLaVA	LLaVA	LLaVA	57.9 (+5.1)	41.2 (+1.9)
Qwen2 LLM	CLIP L/14	Qwen2	LLaVA	LLaVA	LLaVA	63.0 (+5.1)	48.3 (+7.1)
DFN-CLIP encoder	DFN H/14	Qwen2	LLaVA	LLaVA	LLaVA	68.0 (+5.0)	49.5 (+1.2)
Unfreeze LLM	DFN H/14	Qwen2	LLaVA	LLaVA	LLaVA	73.4 (+5.4)	49.9 (+0.4)
Pretrain on PixMo	CLIP L/14	Qwen2	LLaVA	PixMo	LLaVA	66.6	49.4
Molmo-7B-D	CLIP L/14	Qwen2	Molmo	PixMo	Molmo	68.4	58.0
FT Qwen2-VL base on LLaVA	DFN H/14	Qwen2	Qwen2-VL	Qwen2-VL	LLaVA	85.5	61.1
Qwen2-VL-7B	DFN H/14	Qwen2	Qwen2-VL	Qwen2-VL	Qwen2-VL	87.9 (+2.4)	62.4 (+1.3)

4.3.1 PRETRAINING

Setup. We address three questions that are central to understanding the pretraining stage:

- Is a dedicated pretraining stage necessary for LLaVA-architecture models, given that their base components are already pretrained on billion-scale data?
- How does connector-only pretraining compare with jointly unfreezing the LLM?
- To what extent does the quality of the pretraining data govern downstream performance?

To answer these, we compare three pretraining configurations: LLaVA pretraining data (Liu et al., 2023) (a subset of CC-3M with web-scraped captions), Molmo pretraining data (Deitke et al., 2024) (PixMo-Cap, comprising high-quality, human-annotated, detailed captions), and no pretraining. In most configurations we pretrain only the connector, following Liu et al. (2023). We additionally examine a strategy in the spirit of Deitke et al. (2024), pretraining the connector alone for the first 20% of steps as a warmup before unfreezing the LLM for the remaining 80%. Every configuration is followed by one epoch of finetuning on the LLaVA finetuning data.

Findings. Figure 6 reports our pretraining ablation. Connector-only pretraining improves VLM performance in nearly every configuration, the sole exception being the Qwen2 and CLIP ViT-L/14 pairing. The gains are especially pronounced for Vicuna-based VLMs, which improve by an

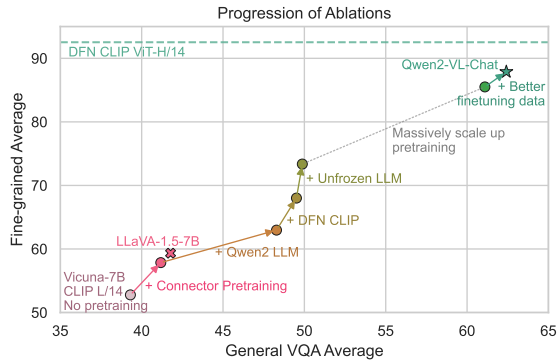


Figure 8: **Cumulative effect of the ablated components.** Beginning from Vicuna-7B with CLIP ViT-L/14, we successively upgrade the base LLM, the vision encoder, and the training settings, raising fine-grained classification accuracy from 52.8% to 73.4%—closing more than half of the gap to Qwen2-VL-Chat’s 87.9%. The residual difference is attributable primarily to Qwen2-VL’s far larger pretraining corpus of 1.4T tokens.

average of +6.1pp on fine-grained classification and +3.1pp on general VQA—confirming that even lightweight connector pretraining confers a substantial, fine-grained-dominant benefit.

Takeaway 3: Large-scale pretraining on image captioning datasets substantially improves fine-grained classification performance but has a more modest effect on general VLM benchmarks.

More strikingly, the quality of the pretraining captions barely registers: pretraining on low-quality web-scraped captions (LLaVA) and on highly detailed human annotations (PixMo) yield almost identical outcomes, with PixMo differing from LLaVA by only -1.6pp on fine-grained performance and +0.4pp on general VQA.

Takeaway 4: Pretraining data quality has a limited impact on overall model performance.

We hypothesize that this insensitivity arises because connector-only pretraining, which leaves the LLM frozen, prevents the model from exploiting the richer signal in higher-quality captions. To test this, we pretrain the LLM jointly with the connector, prefaced by a 20% connector-only warmup. Figure 7 confirms that this joint scheme markedly improves fine-grained classification (+5.5pp) while leaving general VQA scores intact. The improvement materializes under both LLaVA and PixMo pretraining data, however, which further establishes that training on the lower-quality captions does not degrade the LLM’s language capabilities.

Takeaway 5: Pretraining both LLM and connector substantially enhances fine-grained benchmark performance while maintaining general VQA scores.

4.3.2 FINETUNING

Setup. Departing from the preceding protocol, we begin from the pretrained Qwen2-VL-7B-Base model (Wang et al., 2024b)—the model from which Qwen2-VL-7B-Chat is derived—and finetune it on LLaVA-Instruct data (Liu et al., 2023) following our earlier experimental protocol. Comparing the result against Qwen2-VL-7B-Chat, which is instruction-finetuned on 2M in-house examples (Wang et al., 2024b), isolates the effect of the finetuning dataset while holding the base model fixed.

Findings. Finetuning on LLaVA data alone slightly lowers both fine-grained and general performance: fine-grained accuracy falls by 2.4pp (from 87.9% to 85.5%) and general accuracy by 1.3pp (from 62.4% to 61.1%). As Figure 8 makes clear, this is the smallest contribution to fine-grained performance among all our ablations, establishing that instruction finetuning plays a distinctly secondary role in fine-grained knowledge acquisition relative to base model selection and pretraining strategy.

Takeaway 6: The instruction finetuning stage has comparatively less impact on fine-grained classification performance than other factors.

4.4 SUMMARY OF ABLATIONS

Figure 8 and Table 1 synthesize our ablations, tracing how each component contributes to closing the gap between the weakest and strongest VLMs on both fine-grained and general VQA tasks. Starting from the baseline LLaVA architecture (Liu et al., 2023) (Vicuna-7B (Chiang et al., 2023) with CLIP ViT-L/14 (Radford et al., 2021)), the systematic upgrades we introduce raise fine-grained performance from 52.8% to 73.4% and general VQA performance from 39.3% to 49.9%. The decomposition is itself informative: switching to DFN-CLIP (Fang et al., 2023) and unfreezing the LLM during pretraining disproportionately benefit fine-grained capabilities relative to general ability, whereas switching the LLM to Qwen2-7B (Yang et al., 2024) accounts for the majority of the gain in general VQA. The two axes of improvement are thus driven by largely distinct components.

Even after these substantial gains, a 12-point gap in fine-grained classification relative to Qwen2-VL-Chat remains unexplained by our ablations (Table 1). Two candidate factors could account for it: architectural differences and pretraining data scale.

Although architectural differences could in principle contribute, our observational results (Section 3.4) together with our training ablations (Section 4.3) point consistently to pretraining data scale as the dominant factor. Our LLaVA (Liu et al., 2023) and PixMo (Deitke et al., 2024) experiments used comparatively small corpora—each fewer than 1M image-caption pairs, or roughly 200M and 400M tokens, respectively—whereas Wang et al. (2024b) pretrain Qwen2-VL on 1.4T tokens, orders of magnitude more. This disparity in pretraining scale is the most plausible explanation for the residual gap, and it underscores that extensive pretraining is the critical ingredient for developing VLMs with superior fine-grained classification capabilities.

5 LIMITATIONS & CONCLUSION

Our study carries practical limitations that, in turn, delineate concrete opportunities for future work. Under our computational budget we could compare training only at the <1M data-point scale rather than the billion-scale (B) regime employed by some newer VLMs, so the question of how far our conclusions extend to that regime remains open—though the consistent direction of our data-scale evidence suggests they should strengthen, not reverse, with more data. Newer work additionally proposes alternative pretraining strategies whose interaction with fine-grained visual understanding is a natural next target for analysis.

In this work, we systematically evaluate state-of-the-art vision-language models (VLMs) on fine-grained classification benchmarks and establish fine-grained visual classification as a crucial yet underexplored dimension of VLM capability. Through a controlled analysis of the principal model components and training paradigms, we identify which choices govern fine-grained classification and vision-centric competence, yielding concrete, actionable strategies for improving them—and thereby for strengthening the applicability of VLMs in the real-world scenarios that demand precise visual understanding.

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Vision Encoder	LLM	Arch	PT Data	FT Data	Updating	Classification	VLM
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	59.3	41.8
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	59.0	41.2
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	None	LLaVA	Connector	52.8	39.3
CLIP ViT-L/14@336	Llama2-7B	LLaVA	None	LLaVA	Connector	52.5	41.3
DFN ViT-H/14@378	Llama2-7B	LLaVA	None	LLaVA	Connector	56.2	40.0
DFN ViT-H/14@378	Vicuna-7B	LLaVA	None	LLaVA	Connector	52.9	38.2
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	None	LLaVA	Connector	61.4	48.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	None	LLaVA	Connector	66.8	46.0
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	59.7	48.4
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	61.7	43.4
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	LLaVA	Full	85.5	61.1
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	57.9	41.2
DFN ViT-H/14@378	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	61.2	41.4
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	63.0	48.3
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	68.0	49.5
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	63.2	49.9
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	70.2	50.9
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	Qwen2-VL	Full	87.9	62.4
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	57.8	41.6
DFN ViT-H/14@378	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	58.8	43.4
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	60.5	47.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	66.6	49.4
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Full	73.4	49.9
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Full	72.3	49.4

Table 2: **Detailed model-ablation results: aggregate fine-grained classification and general VLM scores.** Each row reports one configuration of vision encoder, LLM, architecture, pretraining data, finetuning data, and weight-updating scheme, alongside its averaged fine-grained classification and VLM accuracy.

Vision Encoder	LLM	Arch	PT Data	FT Data	Updating	MMBench	MMMU	MMStar
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	60.4	32.2	32.7
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	58.6	32.9	32.3
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	None	LLaVA	Connector	54.4	32.8	30.7
CLIP ViT-L/14@336	Llama2-7B	LLaVA	None	LLaVA	Connector	58.0	34.3	31.7
DFN ViT-H/14@378	Llama2-7B	LLaVA	None	LLaVA	Connector	53.9	34.6	31.5
DFN ViT-H/14@378	Vicuna-7B	LLaVA	None	LLaVA	Connector	51.3	33.2	30.1
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	None	LLaVA	Connector	63.1	41.8	40.9
DFN ViT-H/14@378	Qwen2-7B	LLaVA	None	LLaVA	Connector	61.3	36.2	40.5
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	63.2	41.4	40.5
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	55.7	38.3	36.1
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	LLaVA	Full	78.5	49.0	55.8
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	60.4	29.9	33.3
DFN ViT-H/14@378	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	57.4	34.2	32.7
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	64.4	41.3	39.1
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	64.9	41.3	42.3
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	66.5	41.3	41.9
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	68.0	41.6	43.0
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	Qwen2-VL	Full	78.9	49.9	58.4
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	57.8	32.6	34.3
DFN ViT-H/14@378	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	61.1	34.1	34.9
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	64.9	40.0	38.0
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	63.0	41.0	44.3
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Full	66.0	42.0	41.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Full	63.7	40.0	44.6

Table 3: **Detailed model-ablation results on the individual general VQA benchmarks.** Per-benchmark accuracy on MMBench, MMMU, and MMStar for each ablated configuration.

A FULL RESULTS

We report the complete ablation results—aggregated performance, per-benchmark general VQA scores, and per-benchmark fine-grained classification scores—in Table 2, Table 3, and Table 4, respectively.

Vision Encoder	LLM	Arch	PT Data	FT Data	Updating	ImageNet	Flowers	Pets	Food
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	70.9	50.7	46.3	69.4
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	68.7	50.4	46.5	70.3
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	None	LLaVA	Connector	64.3	46.1	40.4	60.4
CLIP ViT-L/14@336	Llama2-7B	LLaVA	None	LLaVA	Connector	64.4	44.2	40.7	60.8
DFN ViT-H/14@378	Llama2-7B	LLaVA	None	LLaVA	Connector	65.4	50.5	46.2	62.8
DFN ViT-H/14@378	Vicuna-7B	LLaVA	None	LLaVA	Connector	62.3	48.7	39.2	61.3
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	None	LLaVA	Connector	70.6	44.1	55.6	75.2
DFN ViT-H/14@378	Qwen2-7B	LLaVA	None	LLaVA	Connector	74.1	50.8	62.3	80.1
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	71.4	39.6	53.1	74.7
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	72.0	46.0	54.1	74.5
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	LLaVA	Full	84.8	79.5	87.1	90.7
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	69.1	50.0	45.3	67.1
DFN ViT-H/14@378	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	72.7	53.2	48.5	70.4
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	71.5	46.9	56.6	76.9
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	76.1	54.1	59.1	82.7
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	74.5	45.8	55.1	77.4
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	78.5	52.3	65.4	84.7
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	Qwen2-VL	Full	85.9	82.3	91.0	92.3
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	69.2	47.4	46.7	67.8
DFN ViT-H/14@378	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	70.8	50.9	44.8	68.5
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	70.8	43.7	53.9	73.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	76.1	50.3	57.8	82.1
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Full	79.4	60.7	67.3	86.0
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Full	80.0	56.4	65.4	87.4

Table 4: **Detailed model-ablation results on the individual fine-grained classification benchmarks.** Per-benchmark accuracy on ImageNet, Flowers, Pets, and Food for each ablated configuration.

B EVALUATION

B.1 FINE-GRAINED CLASSIFICATION

As introduced in Section 3.2, we evaluate VLMs and their base vision encoders on four traditional classification benchmarks, detailed below.

ImageNet-1K (Deng et al., 2009): A foundational image classification dataset organized by the WordNet hierarchy, spanning a broad range of supercategories that each subsume multiple fine-grained subcategories. Its test set comprises 50,000 images distributed over 1,000 classes.

The structure of the ImageNet/WordNet hierarchy, however, leaves many classes too visually distinct from the remainder to admit hard multiple-choice distractors—the “person” supercategory, for instance, contains only three classes. To circumvent this, and the dataset’s unwieldy size, we draw on the human-annotated incorrect model predictions of Recht et al. (2019) to select images for which hard negative answer choices can be constructed. Beginning from 19,056 annotated examples, we distill a pool of 5,457 images covering 928 of the original 1,000 classes. Because this curation leaves the surviving classes unevenly balanced, we report mean-per-class accuracy for ImageNet.

Oxford Flowers-102 (Nilsback & Zisserman, 2008): A dataset of 102 flower species marked by high intra-class variation and pronounced inter-class similarity, with 6,149 test examples.

Oxford-IIIT Pet-37 (Parkhi et al., 2012): A collection of 37 pet categories exhibiting challenging variation in pose, lighting, and occlusion, with 3,669 test examples.

Food-101 (Bossard et al., 2014): A dataset of 101 food categories displaying substantial visual diversity within each class, with 25,250 test examples.

Question generation. For every fine-grained classification test set other than ImageNet (treated above), we construct hard negatives by running zero-shot classification on the test examples with an OpenCLIP ViT-L/14 trained on LAION (Schuhmann et al., 2022), following Geigle et al. (2024). We take the four highest-ranked incorrect predictions and shuffle them together with the correct label to form the answer set.

Prompt formatting. In our multiple-choice VQA format, the question is invariably “What type of object is in this photo?”, and the answer choices are rendered as “A. Option 1\n B. Option 2\n C. Option 3\n D. Option 4\n E. Option 5”. Because different VLMs are trained with different multiple-choice prompt conventions, we adopt each model’s corresponding format from VLMEvalKit (Duan et al., 2024)—for example, “Answer with the option’s letter from the given choices directly.”

For CLIP models we apply the standard zero-shot evaluation procedure, which involves no question format; the class names supplied to the text encoder are instead instantiated in the template “a photo of a {classname}”.

B.2 GENERAL VQA

For the initial observational study of Section 3, we rely directly on the scores reported by the Open-VLM Leaderboard (Duan et al., 2024). By default, that leaderboard reports the average over eight diverse VLM benchmarks: MMBench (Liu et al., 2024b), MMStar (Chen et al., 2024b), MMMU (Val) (Yue et al., 2024), MathVista (Lu et al., 2024), OCRBench (Liu et al., 2024c), AI2D (Kembhavi et al., 2016), HallusionBench (Guan et al., 2024), and MMVet (Yu et al., 2024). This aggregate is used only in Figure 2.

For every subsequent evaluation—of both existing models and the models we train in Section 4—we restrict attention to a subset of multiple-choice VLM benchmarks that permits a more direct, controlled comparison. Specifically, we evaluate on three benchmarks:

MMBench (Liu et al., 2024b): A dataset of 2,948 examples spanning a broad range of VLM capabilities partitioned between perception and reasoning. Most examples target coarse-grained perception and visual reasoning; a subset labeled “fine-grained perception” does exist, but its categories (action recognition, attribute recognition, OCR, and the like) lie in a domain distinct from the fine-grained classification datasets we study.

MMMU (Val) (Yue et al., 2024): A large-scale multimodal dataset spanning a wide distribution of domains and applications. Although the full test set holds 10,500 examples, the field conventionally reports performance on the 900-question validation split, which we adopt. MMMU probes general knowledge tied to visual perception across art, science, engineering, medicine, business, and the humanities.

MMStar (Chen et al., 2024b): A dataset of 1,500 human-selected questions, curated to span a diverse set of tasks across both domain and low-level model capabilities.

Evaluation procedure. For the general VQA benchmarks we defer entirely to the VLMEvalKit evaluation code (Duan et al., 2024). Each existing model carries a predefined evaluation procedure; for our own trained models we apply the LLaVA evaluation code, since we finetune exclusively on LLaVA-Instruct data.

C TRAINING

All experiments build on the LLaVA-1.5 training codebase (Liu et al., 2023), and, except where an ablation explicitly dictates otherwise, adhere to the LLaVA architecture and training procedure.

Pretraining. Concretely, we perform one epoch of pretraining on either the LLaVA pretraining data—image–caption pairs drawn from CC-3M—or the Molmo pretraining data (PixMo-Cap); for PixMo-Cap we use only the “caption” field and discard the raw audio transcript. Pretraining uses a batch size of 256. In most runs we update only the randomly initialized MLP connector, at a learning rate of $1e-3$. When ablating the pretraining strategy, we instead warm up the connector alone for the first 20% of steps and then unfreeze and jointly update the LLM at a learning rate of $2e-5$ for the remaining 80%.

Finetuning. Finetuning always consists of one epoch on the LLaVA-1.5 finetuning data—LLaVA-Instruct together with VQA training data drawn from several datasets to support proper benchmarking. Throughout finetuning we hold the LLM learning rate at $2e-5$ with a batch size of 128, and the vision encoder remains frozen across both training phases.

For the Qwen2-VL-architecture ablation, we initialize from Qwen2-VL-7B Base and train under the same procedure, unfreezing only the LLM.

Compute. All models are trained on 4 nodes of 4 A100 (40GB) GPUs each; a run typically completes within 20 hours, with the precise duration varying with the CLIP model size.

LLaVA vs. PixMo data. The main text repeatedly invokes the disparity in caption quality between the LLaVA and PixMo pretraining data, which we make explicit here. The LLaVA data derives from CC-3M and consists of web-scraped captions notorious for low quality and weak image relevance. PixMo-Cap, by contrast, is collected by having human annotators record a 60–90 second audio description of each image, which is subsequently consolidated into a coherent caption (Deitke et al., 2024). A surface-level comparison quantifies the gap: LLaVA captions average 9.8 words, whereas PixMo captions average 169 words.

LLaVA caption example: “front panel bracket cover for suzuki”

PixMo caption example: “The poster from the TV show ”Law and Order” features a past cast ensemble set in an interrogation room. On the left is a Hispanic man, identified as Detective Nick Amaro, portrayed by Danny Pino. He has dark hair, olive skin, and is dressed in a gray jacket, blue shirt, and black tie. Standing beside him is a blonde woman with shoulder-length hair, who is identified as Detective Amanda Rollins, played by Kelli Giddish. She is ... The backdrop is an interrogation room, with a concrete wall and a portion of a two-way mirror visible, adding to the procedural drama’s atmospheric setting.”